

MARACAY, Venezuela

WMO: 804130

Lat: 10.250N

Lon: 67.649W

Elev: 1338

StdP: 14.00

Time Zone: -4.00 (W04)

Period: 83-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	63.7	65.1	55.3	68.3	83.3	57.4	73.8	81.7	16.9	79.2	14.3	80.7	1.3	50	0.242

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.9	94.7	75.0	93.1	74.6	91.6	74.2	80.2	87.7	79.0	87.1	78.1	86.2	7.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	156.0	84.1	77.1	148.3	82.7	75.9	142.3	81.7	45.1	87.6	43.7	87.6	42.7	86.3	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	13.4	11.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.0	77.5	79.8	80.9	81.3	80.8	78.7	77.6	77.8	78.5	79.1	78.8	77.8
	DBStd	2.50	2.07	2.47	2.63	2.17	2.33	2.26	1.94	2.10	1.71	1.72	1.80	2.05
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10601	852	836	957	940	956	860	855	862	854	902	865	862
	CDD65	5126	387	416	492	490	491	410	390	397	404	437	415	397
	CDH74	45293	3609	4280	5162	5022	4731	3372	2760	2934	3170	3456	3353	3444
	CDH80	18029	1514	1993	2444	2201	1878	1162	865	962	1116	1265	1271	1358
(14) Wind	WSAvg	4.2	4.6	5.1	5.3	4.9	4.5	4.4	3.9	3.7	3.5	3.4	3.5	4.0
(15) Precipitation	PrecAvg	37.0	0.2	0.2	0.3	1.8	3.9	5.1	5.4	6.5	5.4	4.1	2.4	0.6
	PrecMax	62.9	1.4	1.5	2.8	7.8	13.3	16.1	17.9	14.3	10.7	7.9	8.7	3.1
	PrecMin	14.7	0.0	0.0	0.0	0.0	0.3	0.3	1.5	2.4	1.1	1.3	0.1	0.0
	PrecStd	9.6	0.3	0.3	0.6	1.9	2.2	3.0	3.0	2.7	2.5	1.5	1.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.0	96.4	96.8	95.2	94.4	93.1	89.7	90.6	90.8	91.7	91.8	91.5
		MCWB	72.2	74.0	73.4	77.3	75.2	75.7	73.3	75.2	76.3	76.9	76.6	74.6
	2%	DB	91.1	93.6	94.7	93.4	91.8	89.7	87.9	88.4	89.4	89.8	90.0	89.8
		MCWB	71.8	74.4	74.0	75.7	74.2	74.1	73.7	75.3	75.3	75.5	76.0	73.9
	5%	DB	89.5	91.8	92.8	91.6	90.7	88.0	86.4	87.4	87.8	88.1	88.4	88.3
		MCWB	71.4	73.5	74.1	74.6	74.2	73.4	73.6	75.0	75.0	74.9	75.4	73.4
	10%	DB	87.8	89.9	90.6	89.9	88.8	86.2	85.1	85.7	86.2	86.8	87.0	87.4
		MCWB	71.5	72.8	73.9	73.9	74.1	73.5	73.5	74.4	74.7	74.8	75.0	73.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.3	80.1	79.6	81.0	81.0	79.5	78.8	81.3	80.4	80.4	80.4	78.9
		MCDB	86.6	91.8	90.6	91.3	86.9	86.0	85.3	86.6	87.0	86.7	86.7	86.7
	2%	WB	74.6	78.0	77.8	79.0	78.9	77.3	77.0	78.6	78.2	78.2	78.6	76.9
		MCDB	85.6	90.2	89.7	89.8	86.0	84.0	83.4	85.1	84.6	84.9	86.0	86.5
	5%	WB	73.5	76.4	76.5	77.7	77.5	76.0	75.8	77.0	76.9	77.2	77.6	75.4
		MCDB	85.3	87.7	88.6	87.6	84.9	83.1	82.4	83.2	83.5	84.2	84.9	85.1
	10%	WB	72.4	74.9	75.2	76.3	76.5	75.0	74.8	75.6	75.7	76.2	76.4	74.2
		MCDB	84.0	85.7	87.0	85.3	84.0	82.3	81.2	82.0	82.8	83.3	83.6	84.1
(35) Mean Daily Temperature Range	5% DB	MDBR	23.1	22.7	22.1	18.9	16.5	15.8	16.6	16.7	17.3	17.2	18.4	21.0
		MCDBR	24.1	24.1	23.6	20.8	19.2	18.6	18.5	18.4	19.1	18.5	20.0	21.9
		MCWBR	9.3	9.3	8.9	7.8	7.3	7.6	8.4	8.4	8.0	7.6	8.4	9.4
	5% WB	MCDBR	22.8	22.5	21.8	18.6	16.5	15.9	16.8	17.2	17.6	17.3	18.5	20.4
		MCWBR	10.1	10.9	10.0	8.7	8.7	7.9	8.8	9.7	8.6	8.4	9.1	10.0
(40) Clear-Sky Solar Irradiance	taub		0.363	0.386	0.407	0.469	0.444	0.443	0.439	0.409	0.404	0.390	0.377	0.379
	taud		2.567	2.482	2.436	2.246	2.358	2.368	2.375	2.475	2.475	2.534	2.572	2.546
	Ebn at Noon		300	296	290	269	270	267	269	281	286	291	293	292
	Edh at Noon		32	36	38	45	40	39	39	36	36	33	31	32
(44) All-Sky Solar Radiation	RadAvg		1677	1829	1838	1695	1641	1679	1706	1735	1762	1658	1535	1555
	RadStd		68	87	145	128	98	68	65	70	68	54	105	71

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+1.40	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page